Boundary-Layer Meteorol (2008) 129:479–495
DOI 10.1007/s10546-008-9323-9

ORIGINAL PAPER

Measurements and Modelling of the Wind Speed Profile in the Marine Atmospheric Boundary Layer

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Received: 18 December 2007 / Accepted: 18 September 2008 / Published online: 9 October 2008

Abstract We present measurements from 2006 of the marine wind speed profile at a site located 18 km from the west coast of Denmark in the North Sea. Measurements from mast-mounted cup anemometers up to a height of 45 m are extended to 161 m using LiDAR observations. Atmospheric turbulent flux measurements performed in 2004 with a sonic anemometer are compared to a bulk Richardson number formulation of the atmospheric stability. This is used to classify the LiDAR/cup wind speed profiles into atmospheric stability classes. The observations are compared to a simplified model for the wind speed profile that accounts for the effect of the boundary-layer height. For unstable and neutral atmospheric conditions the boundary-layer height could be neglected, whereas for stable conditions it is comparable to the measuring heights and therefore essential to include. It is interesting to note that, although it is derived from a different physical approach, the simplified wind speed profile conforms to the traditional expressions of the surface layer when the effect of the boundary-layer height is neglected.

Keywords Atmospheric stability · Boundary-layer height · Length scales · Marine boundary layer · Wind speed profile

1 Introduction

The knowledge of the wind speed profile in the marine atmospheric boundary layer (ABL) is important because around 70% of the Earth's surface is covered by water. The exchange of momentum, heat, moisture, and CO_2 is essential for global and regional climate, and

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offshore harvesting of wind power has the potential to give a substantial contribution to future energy needs of the society. This means that many of the processes that affect our daily life occur in the atmosphere over the oceans, far away from the human habitat where most of the experimental investigations have been performed. However, our understanding of physical processes in the marine ABL is particularly immature.

The behaviour of the wind speed profile over homogeneous land in stationary conditions has generally been well-predicted within the first 30 to 60 m above ground level by Monin-Obukhov similarity theory (MOST) and surface-layer scaling (Businger et al. 1971; Carl et al. 1973; Tennekes 1973; Högström 1988; Gryning et al. 2007). Beyond these levels it progressively deviates from MOST due to the influence of other scaling parameters such as the boundary-layer height not accounted for in surface-layer scaling (Panofsky 1973; Gryning et al. 2007).

MOST and surface-layer scaling have also been used to predict the behaviour of the wind speed profile over the sea. It has been shown that the observed marine wind speed profile deviates from the MOST profile at heights as low as 29 m above mean sea level (a.m.s.l) (Högström et al. 2006). Lange et al. (2004) found deviations in near-neutral and stable conditions at 50 m a.m.s.l., and Peña and Gryning (2008) at 45 m a.m.s.l. for stable conditions.

It is, however, a challenge to study winds in the marine atmospheric environment. A problem common to most measurements over the sea is related to contamination of observations, to which the wind field measurements are particularly vulnerable. Anchored buoys and meteorological towers offer near ideal measuring conditions with negligible flow deformation but the measuring height is for practical and economical reasons very limited. Observations from ships are economically feasible and practically attractive but suffer from ship motions and site-dependent flow distortion effects. Research and oil platforms are near ideal from a practical point of view but the flow deformation can be immense.

Another problem is the degree to which the measurements represent undisturbed marine conditions. The thermal stratification often changes in coastal regions resulting in large coastal gradients in wind speed and air temperature fields (Niros et al. 2002). Observations of the wind field at meteorological masts located near or at the coast are therefore subject to coastal effects. However, new technologies increase our capabilities of making wind observations in the marine ABL. Satellite scatterometer and synthetic aperture radar ocean wind maps reveal spatial variations in winds at 10 m a.m.s.l. (Christiansen et al. 2006), and platform-based light detection and ranging (LiDAR) allows vertical profiling in the undisturbed atmosphere far above the platform. The LiDAR's performance has been successfully verified over land (Smith et al. 2006; Mann et al. 2007; Kindler et al. 2007) and over sea (Antoniou et al. 2006; Kindler et al. 2007; Peña et al. 2008) by comparison against wind speed observations from mast-mounted cup anemometers.

In this article we develop a simplified model for the marine wind speed profile following the analysis given in Gryning et al. (2007) for the entire ABL over land. The observations are compared to the simplified wind profile model and agree well when the effect of the boundary-layer height is taken into account for the stable atmospheric conditions. This is accomplished using a parameterization proposed by Peña and Gryning (2008) that extracts the relationship between the wind speed profiles and the wind speed-dependent sea roughness length.

Wind speed profile theory is presented in Sect. 2, firstly within the surface layer in Sect. 2.1, and secondly, within the entire ABL in Sect. 2.2. The wind speed profiles are plotted using the parameterization given in Sect. 2.3. Two datasets corresponding to the years 2004 and 2006 are presented in Sect. 3 where the bulk Richardson number is used to classify the measurements according to atmospheric stability. In Sect. 3.2, a comparison is made between the estimation of atmospheric stability using a bulk Richardson number formulation, and that calculated from the turbulent fluxes measured from the sonic anemometer in 2004. The wind shear variation with atmospheric stability is studied in Sect. 4. The observations and models of the wind speed profile are illustrated in Sect. 5. Discussion and concluding remarks are given in the two last sections.

2 Theory for the Wind Profile

Scaling aspects of the wind profile are briefly described following the traditional theory of Businger et al. (1971) for the surface layer, Gryning et al. (2007) for the entire ABL and Peña and Gryning (2008) for the marine ABL. The starting point is the description of the mean wind shear profile for an homogeneous and stationary flow given in Panofsky (1973). Following Gryning et al. (2007) it can be written as:

$$\frac{\partial u}{\partial z} = \frac{u_*}{\kappa \ell} \tag{1}$$

where u is the mean wind speed, z is the height above ground, u_* is the local friction velocity, ℓ is the local length scale, and κ is the von Karman constant ($\approx$0.4).

2.1 Surface Layer

In the surface layer, which accounts for the lowest 10% of the ABL, the variation of u_* with z is neglected, and the length scale is assumed to be equal to the height:

$$u_* = u_{*o}, \tag{2}$$

$$\ell_{SL} = z, \tag{3}$$

where u_{*o} and ℓ_{SL} are the friction velocity and length scale in the surface layer, respectively. The influence of atmospheric stability on ℓ is expressed as:

$$\ell = \ell_{SL} {\phi_m}^{-1} \tag{4}$$

where ϕ_m is the atmospheric stability correction, also known as the dimensionless wind shear, introduced in MOST. Several studies (Businger et al. 1971; Dyer 1974; Högström 1988) suggested power-law expressions for the empirical dependence of ϕ_m on the dimensionless stability parameter, z/L:

$$\phi_m = \left(1 - a\frac{z}{L}\right)^p, \tag{5}$$

$$\phi_m = 1, \tag{6}$$

$$\phi_m = \left(1 + b\frac{z}{L}\right), \tag{7}$$

for unstable, neutral and stable atmospheric conditions, respectively, where L is the Obukhov length given by:

$$L = \frac{-u_*^3 T_o}{g\kappa \overline{w'\Theta_v{}'}}, \tag{8}$$

where T_o corresponds to the mean surface-layer temperature, $\overline{w'\Theta_v{}'}$ is the flux of virtual potential temperature and g is the gravitational acceleration. For stable conditions there is general agreement on the form of ϕ_m. For unstable conditions Businger et al. (1971) suggested $p = -1/4$ based on measurements from the Kansas experiment, whereas Carl et al. (1973) and Grachev et al. (2000) applied $p = -1/3$ that is theoretically preferable because it obeys the free-convection condition.

Introducing Eq. 4 into Eq. 1:

$$\frac{\partial u}{\partial z} = \frac{u_*}{\kappa \ell_{SL}} \phi_m, \tag{9}$$

and replacing ℓ_{SL} in Eq. 9 with Eq. 3, integration with respect to height results in the surface-layer wind profile:

$$u = \frac{u_*}{\kappa} \left[\ln \left(\frac{z}{z_o} \right) - \psi_m \left(\frac{z}{L} \right) \right] \tag{10}$$

where z_o is the aerodynamic roughness length, a theoretical height where u becomes zero near the ground, and ψ_m is a function of the atmospheric stability (Stull 1988).

2.2 Atmospheric Boundary Layer

In the ABL, u_* decreases with z. Here the empirical expression:

$$u_* = u_{*o} \left(1 - \frac{z}{z_i} \right)^{\alpha} \tag{11}$$

is used to account for the slope of the profile and the proximity to the boundary-layer height, z_i. Gryning et al. (2007) gave information about the range of values found in the literature for α and used a value of 1 for simplicity.

In the same study ℓ, in Eq. 1, was not only composed of ℓ_{SL} but of two more length scales that were modelled by inverse summation:

$$\frac{1}{\ell} = \underset{\mathrm{I}}{\frac{1}{\ell_{SL}}} + \underset{\mathrm{II}}{\frac{1}{\ell_{MBL}}} + \underset{\mathrm{III}}{\frac{1}{\ell_{UBL}}} \tag{12}$$

where ℓ_{MBL} and ℓ_{UBL} are the length scales in the middle and upper part of the ABL, respectively. ℓ_{MBL} is not proportional to z (Blackadar 1962; Busch and Panofsky 1968) but varies with atmospheric stability (Gryning et al. 2007), and ℓ_{UBL} is assumed to be equal to the distance to the top of the ABL:

$$\ell_{UBL} = (z_i - z). \tag{13}$$

Under neutral atmospheric conditions, i.e. $\phi_m = 1$, Eqs. 11–13 are inserted into Eq. 1 with $\alpha = 1$ resulting in:

$$\frac{\partial u}{\partial z} = \frac{u_{*o}}{\kappa} \left(1 - \frac{z}{z_i} \right) \left(\frac{1}{z} + \frac{1}{\ell_{MBL}} + \frac{1}{z_i - z} \right). \tag{14}$$

Integration of Eq. 14 with z gives:

$$u = \frac{u_{*o}}{\kappa} \left[\ln \left(\frac{z}{z_o} \right) + \frac{z}{\ell_{MBL}} - \frac{z}{z_i} \left(\frac{z}{2\ell_{MBL}} \right) \right], \tag{15}$$

for $z \gg z_o$.

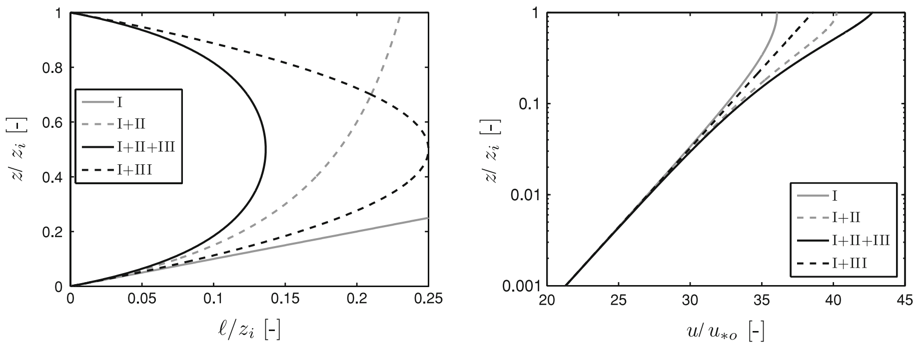

Fig. 1 Profiles of length scale (left) and wind speed (right) for neutral atmospheric conditions with $z_i = 1{,}000$ m, $z_o = 20 \times 10^{-5}$ m and $\ell_{MBL} = 300$ m. The different lines result from the combination of the terms I ($1/\ell_{SL}$), II ($1/\ell_{MBL}$) and III ($1/\ell_{UBL}$) in Eq. 12

Figure 1 illustrates the effect of ℓ_{MBL} on ℓ, and the wind profile for neutral atmospheric conditions. The effect is more pronounced in the middle part of the ABL in such a way that smaller values of ℓ_{MBL} decrease ℓ; a large value of ℓ_{MBL}, corresponding to curve I+III in Fig. 1, implies a logarithmic wind profile in the ABL. Gryning et al. (2007) found, in accordance with the geostrophic drag law, an increase of ℓ_{MBL} for decreasing z_o. It is therefore expected that the smaller z_o over the sea compared to land will result in a wind profile that is closer to logarithmic than those reported for neutral conditions in rural and urban areas by Gryning et al. (2007).

Accounting for the atmospheric stability condition, Gryning et al. (2007) found the following expressions for the wind profile:

$$u = \frac{u_{*o}}{\kappa}\left[\ln\left(\frac{z}{z_o}\right) - \psi_m\left(\frac{z}{L}\right)\left(1 - \frac{z}{2z_i}\right) + \frac{z}{\ell_{MBL}} - \frac{z}{z_i}\left(\frac{z}{2\ell_{MBL}}\right)\right], \tag{16}$$

$$u = \frac{u_{*o}}{\kappa}\left[\ln\left(\frac{z}{z_o}\right) - \psi_m\left(\frac{z}{L}\right) + \frac{z}{\ell_{MBL}} - \frac{z}{z_i}\left(\frac{z}{2\ell_{MBL}}\right)\right], \tag{17}$$

for stable and unstable conditions, respectively. Gryning et al. (2007) showed, based on measurements over land, that ℓ_{MBL} depends on atmospheric stability being smaller in near-neutral conditions and increasing as the atmosphere becomes unstable or stable.

2.3 Wind Profiles Over Water

Peña and Gryning (2008) presented a parameterization of the wind speed profile over the open sea in which the dependency of the sea surface roughness on the wind stress has been extracted. The dependency is expressed by Charnock (1955):

$$z_o = \alpha_c \frac{u_*^2}{g} \tag{18}$$

where α_c is the Charnock's parameter. By using Eq. 18, Peña and Gryning (2008) showed that the surface-layer wind profile, i.e. Eq. 10, can be parameterized as:

$$\frac{u}{u_*} + \underbrace{\frac{1}{\kappa}\ln\left[1 + 2\frac{\Delta u_*}{\overline{u_*}} + \left(\frac{\Delta u_*}{\overline{u_*}}\right)^2\right]}_{\dagger} + \frac{1}{\kappa}\psi_m\left(\frac{z}{\overline{z_o}}\frac{\overline{z_o}}{L}\right) = \frac{1}{\kappa}\ln\left(\frac{z}{\overline{z_o}}\right) \tag{19}$$

where Δu_* is the friction velocity deviation of each wind speed profile from the mean value, $\overline{u_*}$, computed in each stability class, and $\overline{z_o}$ is the mean roughness length defined in the same way as z_o in Eq. 18 but replacing u_* with $\overline{u_*}$. This allows the marine wind speed profile to be studied in a similar way to the wind profile over land with constant roughness by adding the term † to the normalized wind speed in Fig. 1 (right). In this study the measured wind speed profiles are initially divided into atmospheric stability classes, and for each stability class the mean values, $\overline{u_*}$ and $\overline{z_o}$, are derived and used in Eq. 19.

3 Datasets

Two datasets are used in this study:

1. 2004: The measurements are from a meteorological mast (M2) installed in the North Sea near the west coast of Jutland, Denmark, and located at around 18 km from the nearest coastline (Fig. 2). Wind speeds are measured by Risø cup anemometers where three are side-mounted at 15, 30, 45 a.m.s.l. and one top-mounted at 62 a.m.s.l. Wind vanes are placed at 43 and 60 m a.m.s.l. Temperature sensors are available at 55 and 13 m a.m.s.l., and 4 m below mean sea level (b.m.s.l.). A Metek USA-1 sonic anemometer is installed at 50 m a.m.s.l. to measure the turbulence fluxes. The data are available for the months of July to October 2004 and stored as 10-min averages.
   M2 is located at the north-west corner of the Horns Rev wind farm, and wind directions are selected on an "open sea" sector to avoid the influence of the wind farm and the land on the measurements (Peña et al. 2008).
2. 2006: For this year, the same specifications for the wind and temperature data are collected from M2 except that the sonic measurements are missing. In addition, data from a

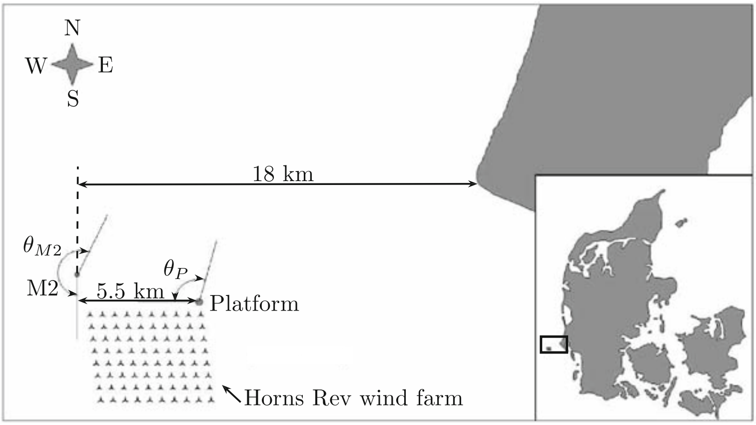

Fig. 2 Location of M2, the platform and the wind farm in the Danish North Sea. θ_P and θ_{M2} correspond to the open sea wind sector observed at the platform and M2 locations, respectively. Each ⅄ symbol indicates the position of a wind turbine in the wind farm

QinetiQ ZephIR® wind LiDAR installed on the transformer/platform of the farm at 20 m a.m.s.l. are available. The LiDAR performed measurements of wind speed at five heights (63, 91, 121, 161 and 300 m a.m.s.l.) where the last one is used for cloud correction (Peña et al. 2008). The description of the LiDAR instrument and the measurement principle is given in Emeis et al. (2007) and Peña et al. (2008). The campaign performed with the LiDAR at Horns Rev is explained in detail in Peña et al. (2008). Both LiDAR and M2 data are available for the months of May to October 2006. The wind directions are selected on the overlapping platform/M2 "open sea" sector ($\theta_P \geq 270° \bigvee \theta_P \leq 10°$).

3.1 Data Processing

Only wind speeds above 3 m s^{-1} at 15 m a.m.s.l. are taken for the analysis. The heat flux observations from the sonic anemometer were corrected for the crosswind contamination using the expression given in Kaimal and Gaynor (1991):

$$\overline{w'T'}_{s(\text{cor})} = \overline{w'T'}_{s(\text{uncor})} + \frac{2u_z\overline{u'w'}}{403} \tag{20}$$

where u_z is the mean wind speed at the sonic height, $\overline{w'T'}_s$ is the sonic sensible heat flux, $\overline{u'w'}$ is the momentum flux, and the constant 403 has units of m^2 s^{-2} K^{-1}.

3.1.1 Bulk Richardson Number

Atmospheric stability from the 2006 data, where the sonic anemometer is not available, is estimated using the bulk Richardson number applied in Grachev and Fairall (1996),

$$Ri_b = -\frac{gz\Delta\Theta_v}{T_z {u_z}^2} \tag{21}$$

where z is the reference height at which the mean temperature T_z and mean wind speed u_z are measured, $\Delta\Theta_v$ is the mean difference between the sea surface temperature (SST) and the virtual potential temperature at the reference height. The Richardson number is related to the dimensionless stability parameter, z/L, depending on the state of the atmosphere:

$$\frac{z}{L} = C_1 Ri_b, \tag{22}$$

for unstable conditions,

$$\frac{z}{L} = \frac{C_2 Ri_b}{1 - C_3 Ri_b}, \tag{23}$$

for $Ri_b < C_3^{-1}$ in stable conditions. Grachev and Fairall (1996) suggest $C_1 = C_2 \approx 10$ and $C_3 \approx 5$, implying a critical $Ri_b \approx 0.2$. It is important to note that the computation of z/L from Eqs. 22 to 23 gives the dimensionless stability parameter as function of the reference height where the bulk Richardson number, Eq. 21, was estimated.

The computation of the bulk Richardson number requires information on the SST that was not available in 2006. Therefore, the sea temperature observation at 4 m b.m.s.l. on M2 is used instead of the SST. By comparing the daily averaged SST observed from a satellite to the observation at 4 m b.m.s.l., it can be seen (Fig. 3) that the differences between both temperatures are small, and both show much less variability than the air temperature. The spatial resolution of the satellite SST is 2 km × 2 km based on observations from NOAA

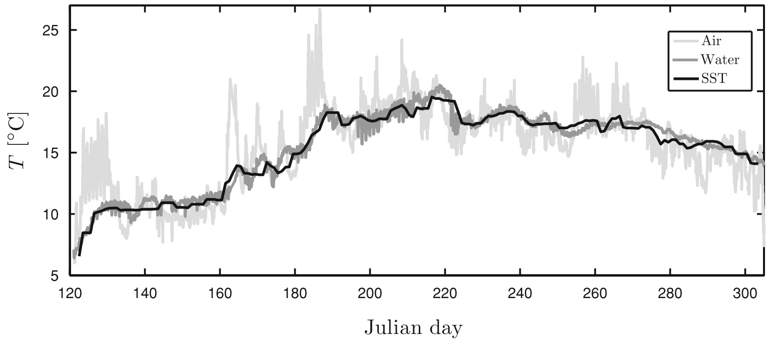

Fig. 3 Comparison of observed temperatures at M2 with satellite SST measurements in 2006. The air temperature is taken at 55 m a.m.s.l. and the water at 4 m b.m.s.l.

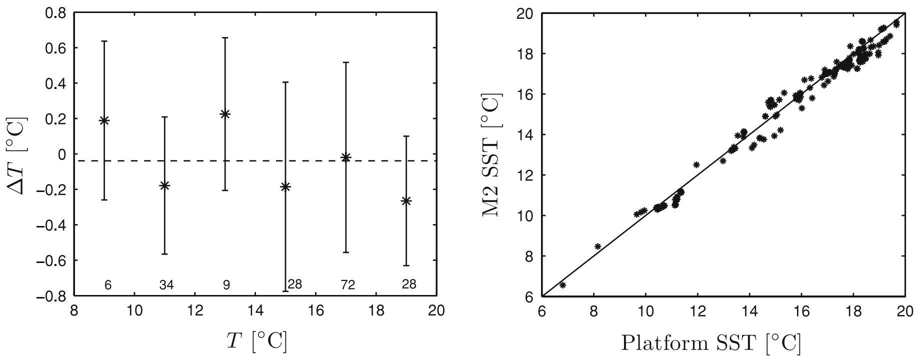

Fig. 4 *To the left*, Temperature difference between the daily average temperature at 4 m b.m.s.l. and the satellite SST for different temperature bins (2°C width). The asterisk represents the mean and error bars the standard deviation within each interval where the number of measurements is also given. *To the right*, Comparison of satellite observations of SST at the locations of the platform and M2

AVHRR 17 and 18, Envisat AATSR, and MODIS Aqua and Terra that are produced from the method described in Høyer and She (2007).

Error statistics on the differences between the satellite SST and the 4 m b.m.s.l. observation for 2°C width temperature bins are shown in Fig. 4 (left). For all temperature bins, the standard deviation is relatively constant and the largest mean value is about 0.2°C. In Fig. 4 (right) the comparison between the satellite SST observations at both M2 and platform locations is illustrated. Due to the lack of suitable temperature measurements at the platform it is assumed, in the analysis of the data from 2006, that the estimated Obukhov length at M2 is also representative at the platform location. Figure 4 (right) reveals that the sea temperature is rather homogeneous between both locations.

3.2 Comparison of Atmospheric Stability from Bulk and Sonic Measurements

The Obukhov length is estimated using both the bulk Richardson number and the sonic anemometer measurements in 2004, and the measurements are classified using the intervals given in Table 1. This classification was suggested in Gryning et al. (2007) from turbulence fluxes observations over land.

Table 1 Stability classes according to Obukhov length

Obukhov length interval (m)	Atmospheric stability class
$10 \leq L \leq 50$	Very stable (vs)
$50 \leq L \leq 200$	Stable (s)
$200 \leq L \leq 500$	Near stable/neutral (ns)
$\|L\| \geq 500$	Neutral (n)
$-500 \leq L \leq -200$	Near unstable/neutral (nu)
$-200 \leq L \leq -100$	Unstable (u)
$-100 \leq L \leq -50$	Very unstable (vu)

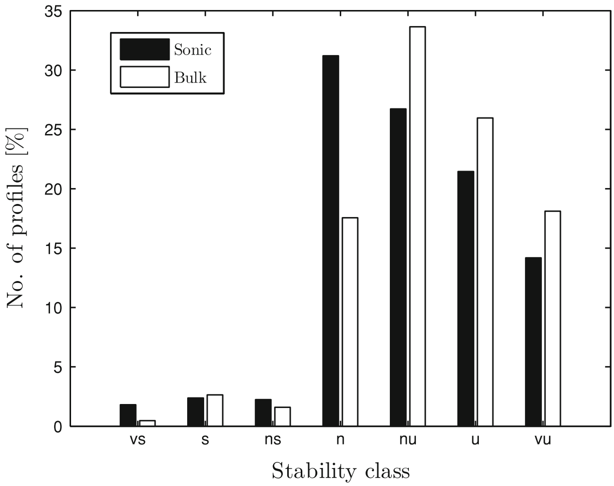

Fig. 5 Availability of wind profiles in different atmospheric stability classes for both sonic and bulk methods in 2004

The values used for the constants in Eqs. 22, 23 are $C_1 = 10$, $C_2 = 10$ and $C_3 = 5$. The bulk Richardson number is computed at a reference height $z = 15$ m a.m.s.l., using the air and water temperature at 13 m a.m.s.l. and 4 m b.m.s.l., respectively, and the wind speed observation at 15 m a.m.s.l. Equation 21 has a contribution from moisture, $\Delta\Theta_v = \Delta\Theta + 0.61 T_z \Delta q$, where Δq is the specific humidity difference between the sea surface and the reference height. Due to lack of humidity observations in 2004, relative humidities of 100% and 80% are assumed at the sea surface and at 13 m a.m.s.l., respectively. These values are estimated from measurements of relative humidity at 13 m a.m.s.l. during a similar period in 2006 at M2 where the mean relative humidity was observed to be 78%. The number of profiles found in each atmospheric stability interval is illustrated in Fig. 5.

The comparison in Fig. 5 shows discrepancies between the Obukhov length derived from the sonic and bulk methods, although the overall agreement is good. One reason for the differences may be that the heat flux observed with the sonic anemometer and used to derive the Obukhov length from Eq. 8 is based on the sonic virtual temperature, therefore, accounting for $\approx 80\%$ of the contribution of the latent heat flux only. Furthermore, the calculation of the bulk Richardson number is carried out with the temperature at 4 m b.m.s.l. instead of the sea surface temperature and a constant relative humidity of 80% is used at 13 m a.m.s.l., thus excluding the effect of humidity fluctuations over the sea. The figure also illustrates the

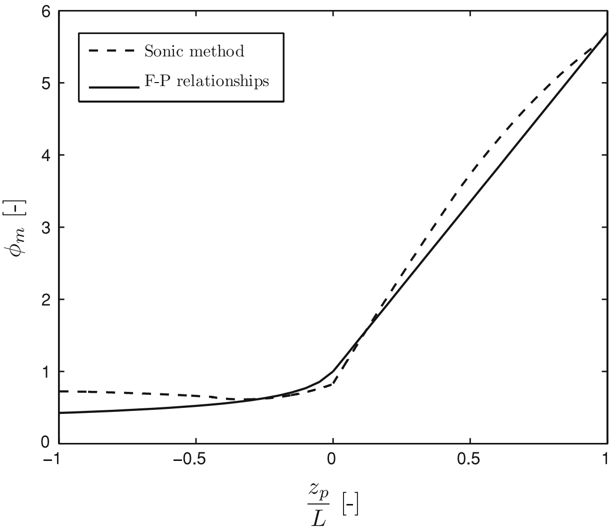

Fig. 6 Variation of the dimensionless wind shear, ϕ_m, with dimensionless stability parameter, z_p/L where $z_p = 27.3$ m, for the sonic method compared with the flux–profile relationships in Eqs. 5–7 using $p = -1/3$, $a = 12$, and $b = 4.7$ in 2004

general difficulties of performing long term measurements in the harsh environment over the sea.

4 Wind Shear

Based on the measurements from 2004 the influence of atmospheric stability on the dimensionless wind shear

$$\phi_m\left(\frac{z}{L}\right) = \frac{\kappa z}{u_*}\frac{\partial u}{\partial z} \tag{24}$$

in the surface layer of the marine ABL is investigated. In order to calculate ϕ_m, information on the wind shear, $\partial u/\partial z$, and u_* is required.

The wind shear is determined applying a method proposed by Högström (1988) (see Appendix) using the wind speed observations at 15, 30 and 45 m a.m.s.l. When sonic anemometer measurements are available both L and u_* can be derived directly (the sonic method).

Figure 6 shows the variation of the ϕ_m function in 2004 estimated from the sonic method compared to the flux–profile relationships in Eqs. 5–7 using $p = -1/3$, $a = 12$, and $b = 4.7$. A common "mean" reference height, $z_p = (45 - 15) / \ln(45/15)$, is used for the calculation of ϕ_m, and z/L is then referenced to this level.

In Fig. 6, a locally weighted curve[1] is fitted to the scatter data in the range $-1 \leq z_p/L \leq 1$ that corresponds to $|L| \geq 27$, and beyond this interval, MOST is not applicable (Webb 1970; Gryning et al. 1987). It can be seen that within an interval around near-neutral conditions, the ϕ_m function derived from the sonic method agrees well with the flux–profile relationships and tends to predict a slightly higher dimensionless wind shear when the stability moves beyond near-neutral conditions.

[1] The locally weighted curve is a least-squares quadratic polynomial fitting.

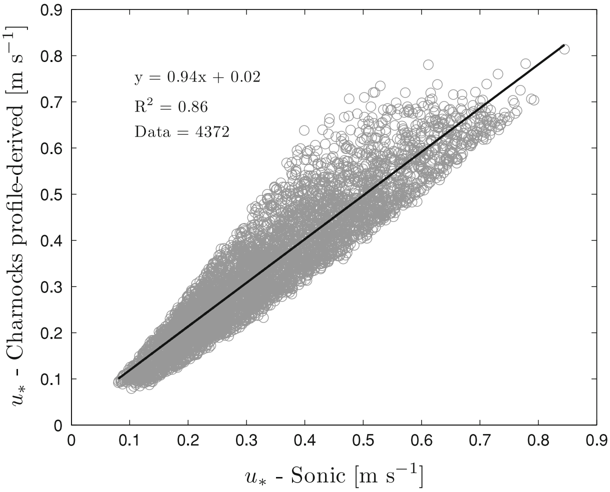

Fig. 7 Comparison of sonic observations of friction velocity and Charnock's profile-derived estimations in 2004 under all atmospheric stabilities

An alternative for the estimation of the surface-layer friction velocity is the use of Charnock's relation, Eq. 18, applying a value $\alpha_c = 1.2 \times 10^{-2}$, in combination with the surface-layer wind profile in Eq. 10 with $p = -1/3$, $a = 12$ and $b = 4.7$ for the computation of the ψ_m function. The ϕ_m function, from which the ψ_m function is derived, was found to predict well the wind shear in comparison with the flux–profile relationships (Fig. 6). This is the so-called Charnock's profile-derived friction velocity:

$$u_* = \frac{u\kappa}{\ln\left(\frac{zg}{\alpha_c {u_*}^2}\right) - \psi_m\left(\frac{z}{L}\right)}, \tag{25}$$

that can be computed from a least-squares fit from measurements of u, where in this case we only use the 15 m a.m.s.l. observation. In Fig. 7 the Charnock's profile-derived friction velocity is compared with the values observed from the sonic anemometer under the whole range of atmospheric stability classes for the 2004 dataset. The Obukhov length derived from the bulk Richardson number, L_b, is used for the estimation of the ψ_m function in Eq. 25. A drag criterion explained in Peña and Gryning (2008) is applied to the data.

Although the correlation is dependent on the amount of data selected with the drag criterion, this is relatively high taking into account that the Charnock's profile-derived value is based on a 15 m a.m.s.l. observation, whereas the sonic measurement is performed at 50 m a.m.s.l.

5 Wind Speed Profiles

The 10-min individual wind speed profiles are classified according to the Obukhov length intervals given in Table 1 and mean wind speed profiles are computed from the observations in 2006. On each mean wind speed profile, the first three heights correspond to cup anemometer observations and the last four to LiDAR observations. A number of mean parameters are computed in Table 2 for each atmospheric stability class where the bulk Richardson number is

Table 2 Computed mean parameters in each atmospheric stability class in 2006

Stability class	$\overline{L_b}$ (m)	$\overline{u_{*o}}$ (m s^{-1})	u_{15} (m s^{-1})	$\overline{z_o}$ (m)	z_i (m)	No. of profiles
vs	28	0.12	4.94	1.9×10^{-5}	122	109
s	85	0.15	5.33	2.9×10^{-5}	150	73
ns	314	0.23	7.08	6.3×10^{-5}	223	18
n	−1531	0.40	11.10	19.6×10^{-5}	393	314
nu	−288	0.42	11.54	22.0×10^{-5}	–	600
u	−139	0.30	8.60	11.1×10^{-5}	–	544
vu	−73	0.22	6.65	6.2×10^{-5}	–	358

used to derive the Obukhov length, L_b, u_{*o} is estimated from the Charnock's profile-derived friction velocity, z_o is derived from Charnock's relation, and the boundary-layer height has been estimated by the diagnostic expression (Rossby and Montgomery 1935):

$$z_i = 0.12\frac{\overline{u_*}}{|f_c|} \tag{26}$$

where f_c is the Coriolis parameter. This is applied to the stable and neutral classes only because of the lack of suitable diagnostic expressions for the height of the unstable ABL (Gryning and Batchvarova 2002).

From Table 2 it is observed that the highest values computed for $\overline{u_*}$ and u are found when atmospheric conditions are close to neutral in agreement with Peña and Gryning (2008). It is also interesting to note that the computation of the mean drag coefficient for neutral conditions, $C_z = (u_*/u_z)^2$, using the values in Table 2 gives $\overline{C_{15}} = 1.3 \times 10^{-3}$, and equal to the weighted mean found by Kraus (1972), $\overline{C_{10}} = 1.3 \times 10^{-3}$, based on near-neutral observations representing open sea conditions. Extrapolating the mean wind speed at 15 to 10 m a.m.s.l. using Eq. 10 and the neutral data in Table 2, $u_{10} = 10.70\,\mathrm{m\,s^{-1}}$, and applying the power-law relation, $C_{10} \times 10^3 = 0.51{u_{10}}^{0.46}$, given in Garratt (1977), the drag coefficient results in $C_{10} = 1.52 \times 10^{-3}$, then $C_{15} < C_{10}$ as expected.

It is not difficult to demonstrate that when $\ell_{MBL} \gg z$ in Eqs. 15–17, the wind speed profile is given by:

$$u = \frac{u_{*o}}{\kappa}\ln\left(\frac{z}{z_o}\right), \tag{27}$$

$$u = \frac{u_{*o}}{\kappa}\left[\ln\left(\frac{z}{z_o}\right) - \psi_m\left(\frac{z}{L}\right)\left(1-\frac{z}{2z_i}\right)\right], \tag{28}$$

$$u = \frac{u_{*o}}{\kappa}\left[\ln\left(\frac{z}{z_o}\right) - \psi_m\left(\frac{z}{L}\right)\right] \tag{29}$$

for neutral, stable and unstable atmospheric conditions, respectively. These expressions are equivalent to the surface-layer wind profile given in Eq. 10 under neutral and unstable conditions, although the physical assumptions are different. In the case of stable conditions, a correction due to the proximity of the boundary-layer height is still present.

The comparison between the observations and the predicted profiles, Eqs. 27–29, is illustrated in Figs. 8, 9 where z_o is derived from Eq. 18 and the influence of z_i is neglected when using Eq. 28 in Fig. 8. The curves are plotted using Charnock's non-dimensional profiles

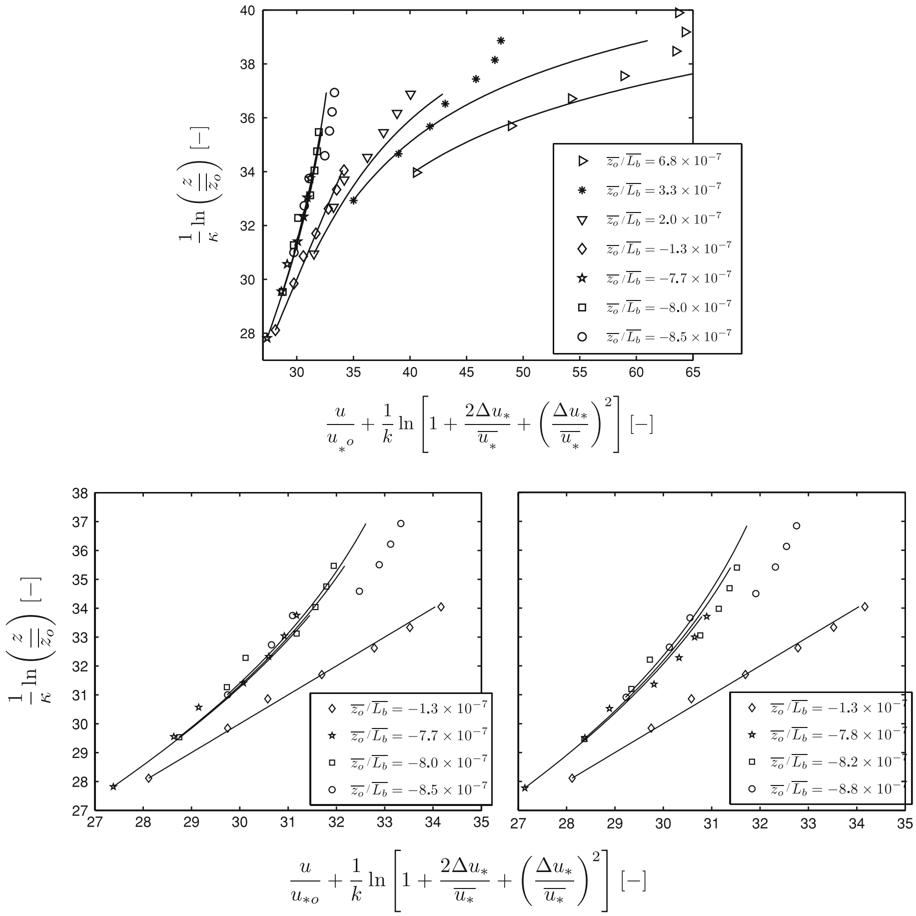

Fig. 8 Comparison of measurements (markers) and non-dimensional wind speed profiles (solid lines) for each atmospheric stability class. The effect of z_i is neglected. The lower panels show the unstable and neutral wind profiles within a much smaller range of dimensionless wind speeds: $a = 12$ (left) and $a = 19$ (right)

given in Peña and Gryning (2008). The differences in the dimensionless height range are due to $\overline{z_o}$ values that increase as the atmosphere approaches neutral conditions (Table 2). Thus, the neutral and near unstable/neutral profile lie within the smallest dimensionless height ranges.

The curves in Fig. 8 show a special feature of the non-dimensional stability, $\overline{z_o}/L$: in the upper panel the three unstable curves are indistinguishable because they have approximately the same $\overline{z_o}/L$, although the individual values for $\overline{z_o}$ and $\overline{L}$ are different (Table 2). They correspond to the observations with star and square markers in the lower panels where only two unstable curves are indistinguishable due to the re-scaled horizontal axis.

As Fig. 8 (upper panel) shows, the LiDAR observations match well with the cup anemometer profiles for stable atmospheric stability classes. The two lower panels in Fig. 8 show in detail the wind speed profiles on the unstable-neutral side, where the LiDAR observations are found to have poorer matching with the mast profiles compared to stable conditions. Both LiDAR and mast measurements show the same profile shape but the LiDAR gives a higher wind speed.

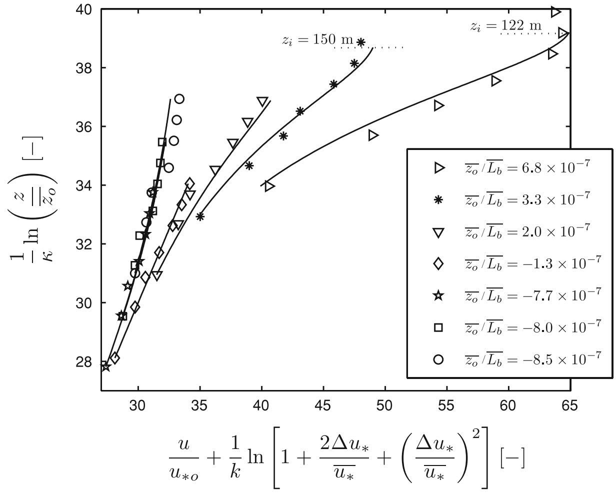

Fig. 9 Comparison of measurements (markers) and non-dimensional wind speed profiles (solid lines) for each atmospheric stability class adding the expression for the boundary-layer height on the stable prediction

The mast profiles are well predicted by the theoretical curves on the unstable side. The lower panel to the left in Fig. 8 is obtained with the traditional value $a = 12$, whereas a better fit is obtained when the analysis is performed with a value $a = 19$ in the lower panel to the right.

An interesting result from Fig. 8 (upper panel) is the deviation of the observations from the stable mean wind profiles—the three datasets and curves at the right part. This might be due to the influence of the boundary-layer height, and in Fig. 9 the values of z_i given in Table 2 are taken into account for the stable wind profiles.

There is naturally no change on the neutral and unstable profiles. For the stable profiles, the correction due to the boundary-layer height improves the agreement with the observations. The formulation of the stable wind profile is valid until the boundary-layer height is reached, as indicated in Fig. 9 where the boundary-layer height is shown in terms of the dimensionless height. For the near stable/neutral profiles, the computed boundary-layer height is beyond the height of the observations, whereas it is within the measurement range in the stable and very stable profiles.

6 Discussion

Although they are derived from a different physical approach, the simplified expressions for the neutral and unstable wind profiles, Eqs. 27 and 29, conform to the traditional expression for the surface-layer wind profile in Eq. 10 where MOST is used to correct the mean wind shear in diabatic conditions. The approach taken in this study is to apply MOST to the surface-layer length scale only, resulting in Eqs. 15–17. This leads to simplified expressions for the marine wind speed profile, which has good agreement with wind speed observations up to 161 m a.m.s.l.

The Obukhov length, and consequently the atmospheric stability, takes into account the effect of moisture fluctuations on buoyancy. The sonic anemometer measurements account for approximately 80% of the contribution of the moisture fluctuations to the virtual temperature, and therefore the moisture flux is only partly accounted for. The estimate of the Obukhov length derived from the bulk Richardson number is based on temperature differences and on an assumption of a mean relative humidity of 80% at 15 m a.m.s.l., because measurements of the humidity were not available. Despite these shortcomings in the measurements, the contribution from humidity fluctuations is important especially close to neutral stability conditions.

The height of the neutral and stable boundary layer was estimated from Eq. 26, which takes into account the mechanical contribution of turbulence only. The relation is here used for stable conditions as well, because of the abundance of diagnostic parameterizations of the height of the stable boundary layer, based on different parameters and providing very different estimates. Presently there is no general agreement on a preference for any of the diagnostic parameterizations and Eq. 26 was therefore used for simplicity due to the lack of measurements of the boundary-layer height over the sea. An overview of such parameterizations can be found in Seibert et al. (2000).

7 Conclusion

The study of the wind speed profile over the sea is performed using the so-called Charnock's non-dimensionless wind profiles where the dependency of the sea roughness length on the wind speed is extracted. This has been found to be a useful tool for analysing a large dataset of marine wind speed profiles for open sea conditions.

Expressions for the marine ABL wind speed profile are compared with combined LiDAR/cup observations up to 161 m a.m.s.l. When not accounting for the boundary-layer height the prediction with a simplified wind-speed-profile expression is good under unstable and neutral conditions, but overpredicts the wind speed under stable conditions at heights above 30–40 m a.m.s.l. By adding a term involving the boundary-layer height, this overprediction is well corrected up to the boundary-layer height, which is estimated to be within the observation range of the wind speed profile for stable and very stable atmospheric conditions.

The method to derive the Obukhov length from the bulk Richardson number has been compared to sonic data from 2004. It has been demonstrated that the surface-layer friction velocity can be estimated from the surface-layer mean wind speed profile in combination with Charnock's relation by comparison to the estimation of friction velocity from the sonic anemometer observations. The variation of the ϕ_m function with z/L from the sonic anemometer measurements agrees well, within an interval close to neutral conditions, with the suggested flux–profile relationships given in Businger et al. (1971) and Dyer (1974).

The measured wind speed profiles are compared to the parameterization suggested for land conditions by Gryning et al. (2007). The best agreement is found by neglecting the length scale, ℓ_{MBL}, for the middle part of the ABL, and this can be argued to be a consequence of the low roughness length over the sea. Neglecting ℓ_{MBL} retains the basic shape of the length scale profile suggested in Gryning et al. (2007). Whether this is a site specific feature or more general for the marine ABL is not clear.

Acknowledgements The authors want to thank: Kurt Hansen from the Technical University of Denmark for making available the measurements corresponding to 2004, Jacob L. Høyer from the Center for Ocean and Ice, DMI, for the SST analysis, Lotte Sørensen, Søren Larsen and Hans Jørgensen from Risø DTU and Ekaterina Batchvarova from the National Institute of Meteorology and Hydrology in Bulgaria for constructive comments on the manuscript, Claire Vincent from Risø DTU for advice on English language, Michael Courtney from Risø DTU for the continuous help acquiring the LiDAR data, and an anonymous reviewer for the comments that improved considerably the original manuscript. AP would like to thank Henrik Søgaard from the

University of Copenhagen for the supervision of my PhD. The work has been carried out in collaboration with Paul Sørensen from DONG energy. Funding from The Danish Council for Strategic Research to the project "12 MW" Sagsnr. 2104-05-0013 is also acknowledged. The work is related to activities within COST728 and COST735.

Appendix: Estimation of the ϕ_m Function

The dimensionless wind shear in Eq. 24 can be computed from mean wind speed measurements taken at different heights. Therefore, the assumption $\partial u/\partial z \approx \Delta u/\Delta z$ is commonly made. A better estimate of ϕ_m is performed when wind speed measurements at several heights are taken into account. Högström (1988) approximated the wind speed profile by a second-order polynomial in $\ln(z)$:

$$u = u_o + A\ln(z) + B[\ln(z)^2], \tag{30}$$

where the terms u_o, A and B are determined by a least-squares method. Differentiation of Eq. 30 with z gives:

$$\frac{\partial u}{\partial z} = \frac{A + 2B\ln(z)}{z}, \tag{31}$$

and introducing Eq. 31 into Eq. 24, gives

$$\phi_m = \frac{\kappa}{u_*}\left[A + 2B\ln(z)\right]. \tag{32}$$

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